



EGH404-RESEARCH IN ENGINEERING PRACTICE

INTRODUCTION TO CORRELATION

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Correlation - Introduction

Used to understand the relationship between variables.

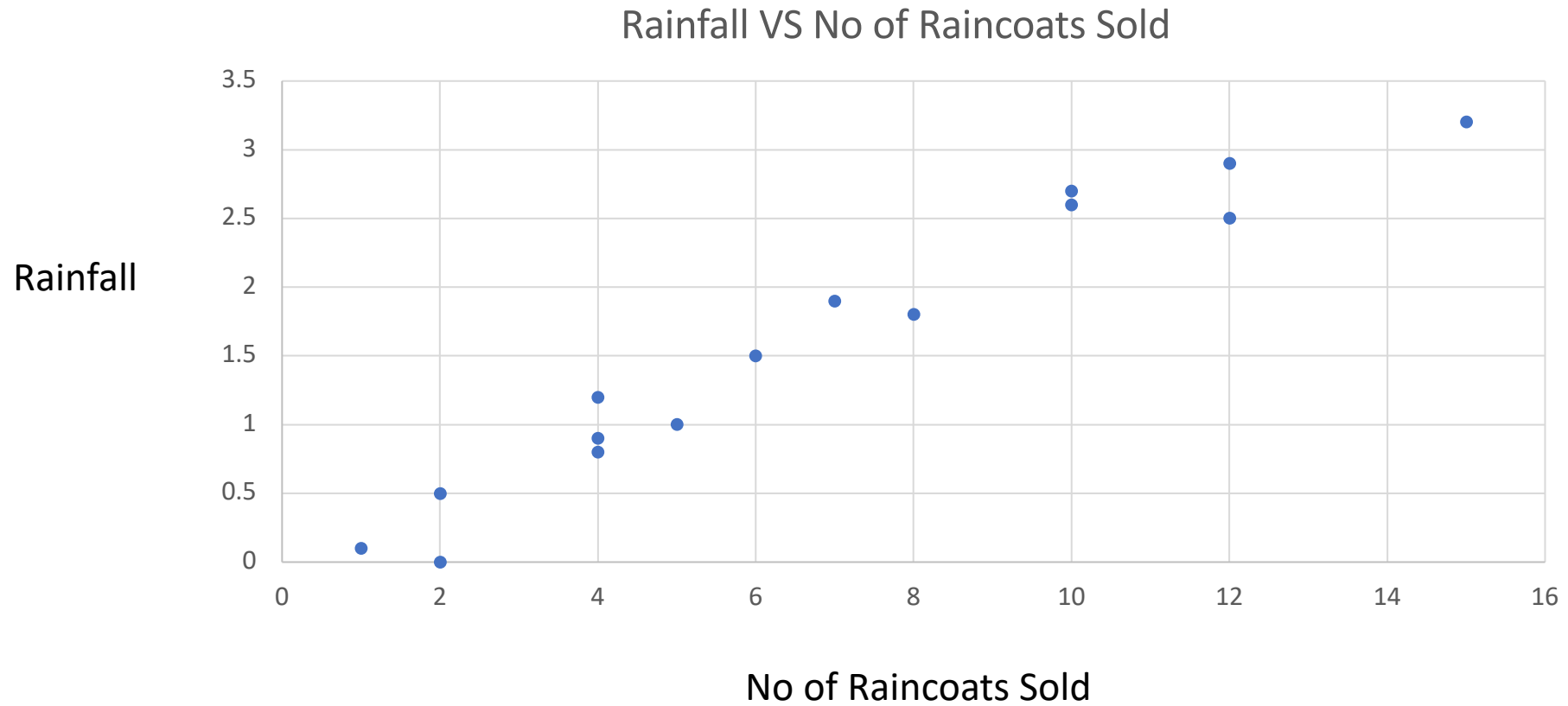


RAIN



RAINCOAT

Correlation - Introduction

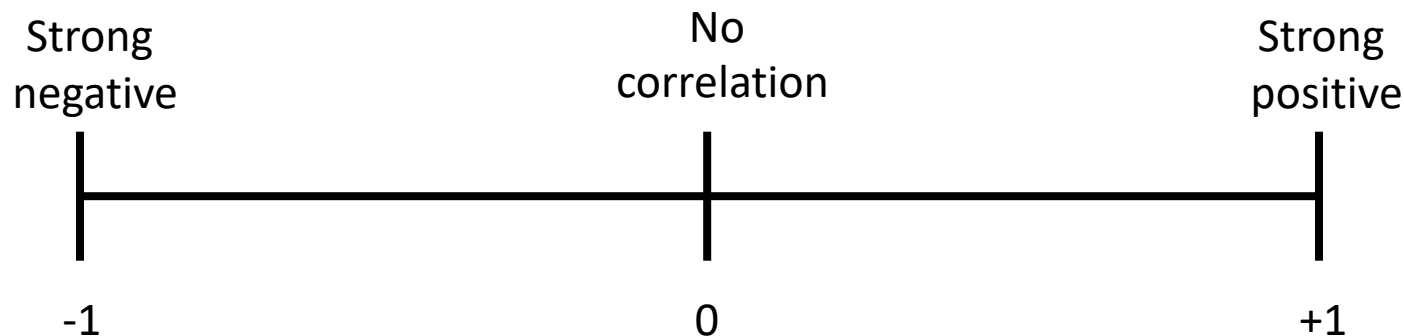


Correlation - Properties

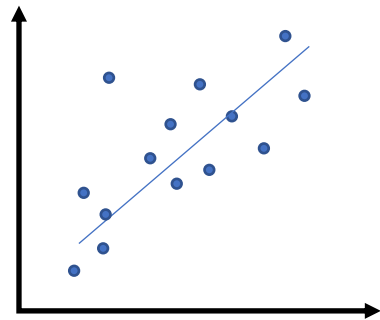
- **Direction:** Correlation can be positive (both variables move in the same direction) or negative (one variable moves up while the other moves down).
- **Strength:** The strength of the relationship is measured on a scale from -1 to 1. A correlation of 1 indicates a perfect positive relationship, -1 indicates a perfect negative relationship, and 0 indicates no relationship.

Correlation Coefficient (r)

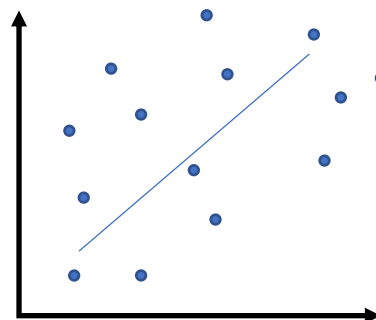
- The correlation coefficient quantifies the degree of correlation. It ranges from -1 to 1, with values closer to the extremes indicating stronger relationships.
- Here we are discussing Pearson correlation coefficient: Measures linear correlation between two variables.



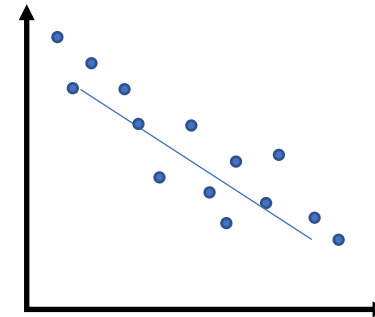
Correlation Coefficient (r)



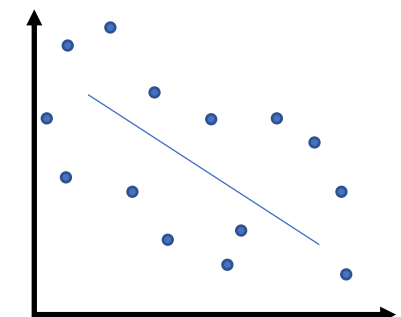
$r = 0.8$



$r = 0.3$



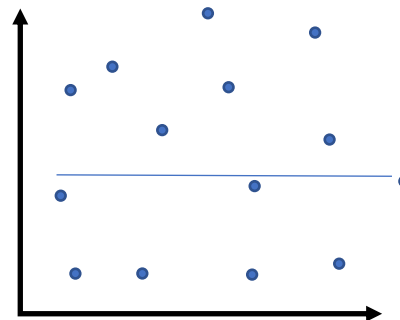
$r = -0.9$



$r = -0.2$

Positive

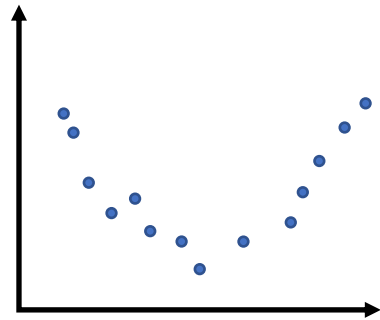
Negative



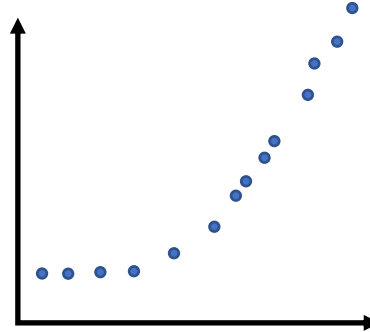
$r = 0.0$

No relationship

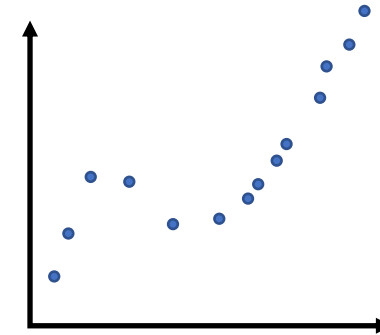
Types of Relationships Among Variables



Quadratic



Exponential

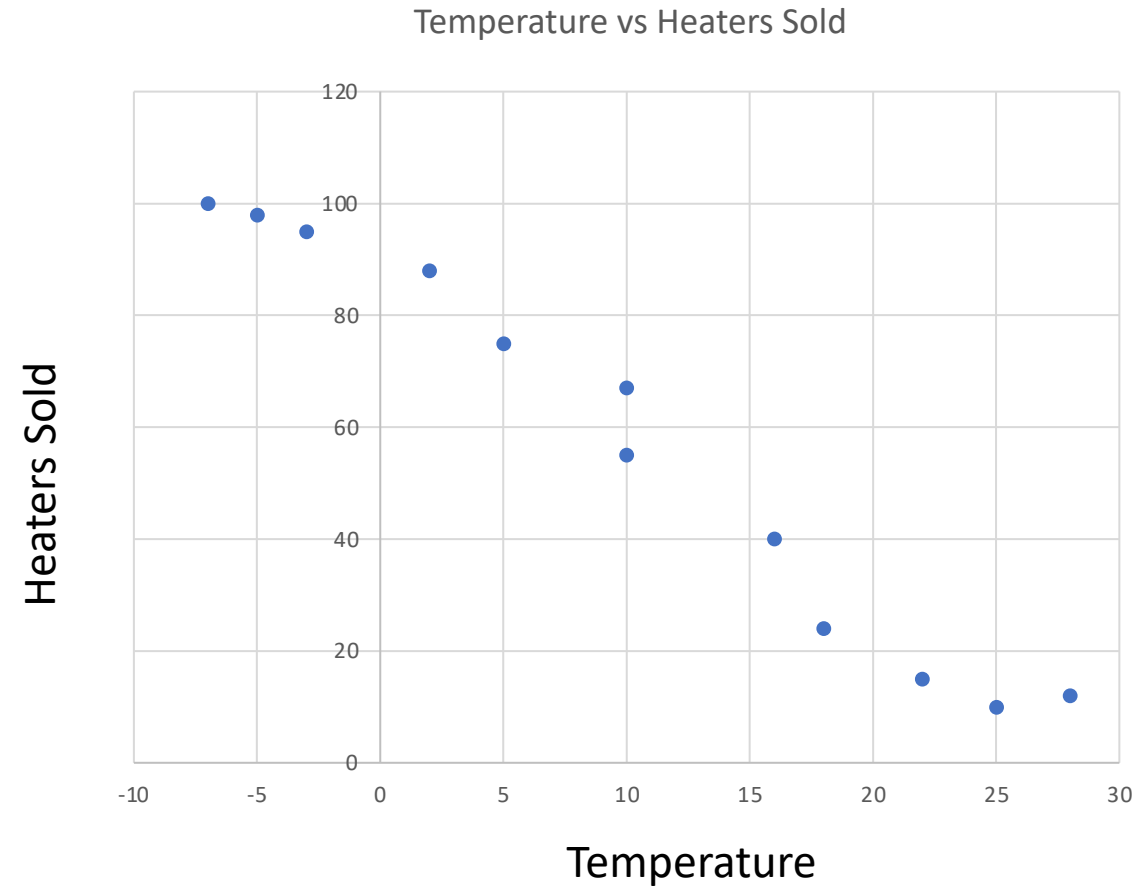


Polynomial

Always use scatter plot and first plot your data to see whether the the relation ship is linear

Example in Excel

Month	Temperature	Heaters Sold
Jan	-5	98
Feb	-7	100
Mar	5	75
Apr	10	67
May	18	24
Jun	22	15
Jul	28	12
Aug	25	10
Sep	16	40
Oct	10	55
Nov	2	88
Dec	-3	95



Example in Excel

F1	=CORREL(Table1[[Temperature]],Table1[Heaters Sold])					
	A	B	C	D	E	F
1	Month	Temperature	Heaters Sold		Correlation Coefficient	-0.9840577
2	Jan	-5	98			
3	Feb	-7	100			
4	Mar	5	75			
5	Apr	10	67			
6	May	18	24			
7	Jun	22	15			
8	Jul	28	12			
9	Aug	25	10			
10	Sep	16	40			
11	Oct	10	55			
12	Nov	2	88			
13	Dec	-3	95			

Correlation Matrix

Civil Engineering: Cement Manufacturing Dataset

<https://www.kaggle.com/datasets/vinayakshanawad/cement-manufacturing-concrete-dataset>

cement	slag	ash	water	strength
141.3	212	0	203.5	29.89
168.9	42.2	124.3	158.3	23.51
250	0	95.7	187.4	29.22
266	114	0	228	45.85
154.8	183.4	0	193.3	18.29
255	0	0	192	21.86
166.8	250.2	0	203.5	15.75
251.4	0	118.3	188.5	36.64
296	0	0	192	21.65
155	184	143	194	28.99
151.8	178.1	138.7	167.5	36.35
173	116	0	192	6.94
385	0	0	186	27.92

Correlation Matrix

	cement	slag	ash	water	strength
cement		-0.2752	-0.3975	-0.0816	0.4978
slag	-0.2752		-0.3236	0.10725	0.1348
ash	-0.3975	-0.3236		-0.257	-0.106
water	-0.0816	0.10725	-0.257		-0.29

Correlation Formulation

$$r = \frac{\textit{covariance} (x, y)}{\textit{standard Deviation} (x) \times \textit{standard Deviation} (y)}$$

$$r = \frac{\sum (x_i - x_{\text{average}}) (y_i - y_{\text{average}})}{\sqrt{\sum (x_i - x_{\text{average}})^2 * \sum (y_i - y_{\text{average}})^2}}$$

Correlation VS Causation

Correlation $\neq$ Causation

Correlation VS Causation

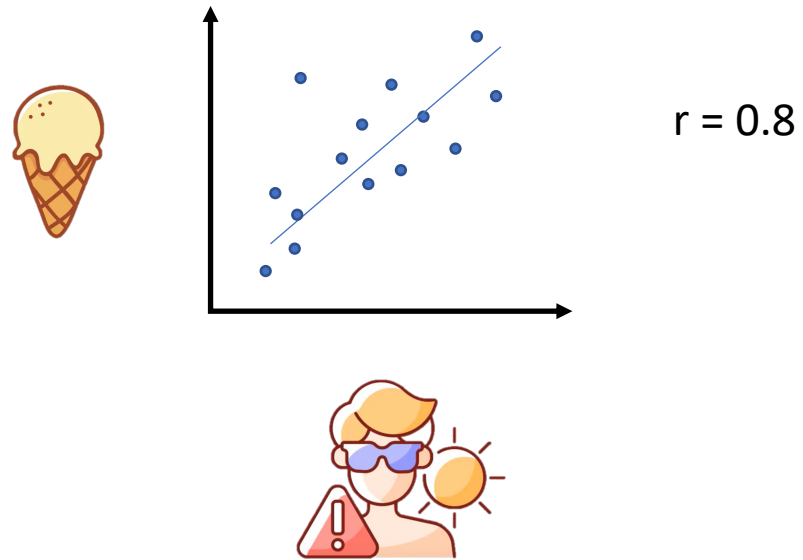


Ice Cream Sales



Sunburns

Correlation VS Causation



Correlation VS Causation

